

1.

A stable LTI system is characterized by the following difference equation:

$$\frac{1}{2}y[n-1] - \frac{9}{4}y[n] + y[n+1] = x[n]$$

- Calculate the system function  $H(z)$ .
- Calculate the impulse response  $h[n]$ .

Formlerne kan findes i bogen (second edition)  
S. 245-246 eller i giti's slides (Begynder på slide 7)

a)

$$\sum_{k=0}^N a_k y[n-k] = \sum_{m=0}^M b_m x[n-m]$$

$$\Downarrow$$

$$\sum_{k=0}^N a_k z^k Y(z) = \sum_{m=0}^M b_m z^m X(z)$$

$$\Downarrow$$

$$\left( \sum_{k=0}^N a_k z^k \right) Y(z) = \left( \sum_{m=0}^M b_m z^m \right) X(z)$$

$$\Downarrow$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{m=0}^M b_m z^m}{\sum_{k=0}^N a_k z^k}$$

$$\frac{1}{2}y[n-1] - \frac{9}{4}y[n] + y[n+1] = x[n]$$

$$\Downarrow$$

$$\frac{1}{2}z^{-1}Y(z) - \frac{9}{4}Y(z) + z^1Y(z) = z^0X(z)$$

$$\Downarrow$$

$$\left( \frac{1}{2}z^{-1} - \frac{9}{4} + z^1 \right) Y(z) = (z^0) X(z)$$

$$\Downarrow$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{\sum_{m=0}^M b_m z^m}{\sum_{k=0}^N a_k z^k}$$

$$\frac{Y(z)}{X(z)} = \frac{1}{\left( \frac{1}{2}z^{-1} - \frac{9}{4} + z^1 \right)} \xrightarrow{\text{ganger } z^1 \text{ igennem}} \frac{z^1}{\left( \frac{1}{2} - \frac{9}{4}z + z^2 \right)}$$

Split op for at lave  
Partial fraction efter:

$$p - \frac{9}{4}z + z^2 \rightarrow (z+a)(z+b)$$

$$q = a+b = -\frac{9}{4}$$

$$p = a \cdot b = \frac{1}{2}$$

$$\text{Solve } (\{q, p\}, \{a, b\}) = \left\{ a = -\frac{1}{4}, b = -2 \right\}, \left\{ a = -2, b = -\frac{1}{4} \right\}$$

$$(z+a)(z+b) = (z - \frac{1}{4})(z - 2)$$

$$H(z) = \frac{z^{-1}}{(z - \frac{1}{4})(z - 2)}$$

$$\frac{A_1}{(z - \frac{1}{4})} + \frac{A_2}{(z - 2)}$$

$$z - \frac{1}{4} = 0 \Rightarrow z = \frac{1}{4} \quad z = a_k$$

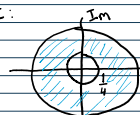
$$z - 2 = 0 \Rightarrow z = 2$$

$$A_1 = (z - \frac{1}{4}) \cdot X(z) \Big|_{z=\frac{1}{4}} = (z - \frac{1}{4}) \cdot \frac{z^{-1}}{(z - \frac{1}{4})(z - 2)} \Big|_{z=\frac{1}{4}} = -\frac{1}{2}$$

$$A_2 = (z - 2) \cdot X(z) \Big|_{z=2} = (z - 2) \cdot \frac{z^{-1}}{(z - \frac{1}{4})(z - 2)} \Big|_{z=2} = \frac{8}{7}$$

$$X(z) = \frac{1}{z} = \frac{1}{(z - \frac{1}{4})} + \frac{8}{(z - 2)} \xrightarrow{\text{ganger } z^{-1} \text{ igennem så tabelopslag er nemmere}} \frac{-\frac{1}{2}z^{-1}}{(1 - \frac{1}{4}z^{-1})} + \frac{\frac{8}{7}z^{-1}}{(1 - 2z^{-1})}$$

ROC:

If system is  
stable:

$$\frac{1}{4} < |z| < 2$$

b)

APPLY inverse z

Right-sided

Left-sided

$$\frac{-\frac{1}{2}z^{-1}}{(1 - \frac{1}{4}z^{-1})} + \frac{\frac{8}{7}z^{-1}}{(1 - 2z^{-1})}$$

$$\frac{1}{4} < |z| < 2$$

$$\left( z^{-1} \right) \cdot \left( \frac{1}{4} \right)^{n-1}$$

$$-\frac{1}{2} \cdot \left( \frac{1}{4} \right)^{n-1} u[n-1] - \frac{8}{7} \cdot 2^{n-1} \cdot \underbrace{u[n-1-1]}_{u[n-2]}$$

$$2) \quad x[n] = \left(\frac{1}{2}\right)^n u[n] + (2)^n u[-n-1]$$

a) By inspection, the z-transform of the input above is

$$X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} - \frac{1}{1 - 2z^{-1}} \quad \begin{matrix} |z| > \frac{1}{2} \\ |z| < 2 \end{matrix} \Rightarrow \frac{1}{2} < |z| < 2$$

$$y[n] = 6\left(\frac{1}{2}\right)^n u[n] - 6\left(\frac{3}{4}\right)^n u[n]$$

By inspection, the z-transform of the output signal is given by

$$Y(z) = \frac{6}{1 - \frac{1}{2}z^{-1}} - \frac{6}{1 - \frac{3}{4}z^{-1}} \quad \begin{matrix} |z| > \frac{1}{2} \\ |z| > \frac{3}{4} \end{matrix} \rightarrow |z| > \frac{3}{4}$$

Therefore:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{6 - \frac{3}{2}z^{-1} - 6 + 3z^{-1}}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - \frac{3}{4}z^{-1}\right)} \cdot \frac{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - 2z^{-1}\right)}{1 - 2z^{-1} - 1 + \frac{1}{2}z^{-1}} = \frac{-\frac{3}{2}z^{-1}}{\left(1 - \frac{3}{4}z^{-1}\right) \cdot -\frac{3}{2}z^{-1}} = \frac{1 - 2z^{-1}}{1 - \frac{3}{4}z^{-1}}, \quad |z| > \frac{3}{4}$$

Han ganger bare på kryds så det kan komme på  $H(z) = \frac{Y(z)}{X(z)}$

$$3) \quad x[n] = -\frac{1}{3}\left(\frac{1}{2}\right)^n u[n] - \frac{1}{3}(2)^n u[-n-1]$$

By inspection, the z-transform of  $x[n]$  is

$$X(z) = -\frac{1}{3}\left(\frac{1}{1 - \frac{1}{2}z^{-1}}\right) + \frac{1}{3}\left(\frac{1}{1 - 2z^{-1}}\right) \quad \begin{matrix} |z| > \frac{1}{2} \\ |z| < 2 \end{matrix} \Rightarrow \frac{1}{2} < |z| < 2$$

$$X(z) = \frac{-\frac{1}{3}(1 - 2z^{-1}) + \frac{1}{3}(1 - \frac{1}{2}z^{-1})}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - 2z^{-1}\right)} = \frac{1}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - 2z^{-1}\right)}$$

$$Y(z) = \frac{1 - z^{-2}}{\left(1 - \frac{1}{2}z^{-1}\right)\left(1 - 2z^{-1}\right)}$$

$Y(z)$  has the same poles as  $X(z) \rightarrow \frac{1}{2} < |z| < 2$

$$H(z) = \frac{Y(z)}{X(z)} = 1 - z^{-2} \Rightarrow h[n] = \delta[n] - \delta[n-2]$$

$$4) \quad H(z) = \frac{1 - 3z^{-1}}{(1 - z^{-1})(1 - 2z^{-1})} \quad |z| > 2$$

$$H_i(z) = \frac{(1 - z^{-1})(1 - 2z^{-1})}{1 - 3z^{-1}} = \frac{1 - 3z^{-1} + 2z^{-2}}{1 - 3z^{-1}}$$

$$\begin{array}{r} -\frac{2}{3}z^{-1} + \frac{7}{9} \\ 2z^2 - 3z^{-1} + 1 \\ \hline 2z^2 - \frac{2}{3}z^{-1} \\ \hline \text{" } -\frac{7}{3}z^{-1} + 1 \\ -\frac{7}{3}z^{-1} + \frac{7}{9} \\ \hline \text{" } \frac{2}{9} \end{array}$$

$$\rightarrow H_i(z) = \frac{7}{9} - \frac{2}{3}z^{-1} + \frac{2}{9} \cdot \frac{1}{1 - 3z^{-1}}$$

The possible choices for the region of convergence of  $H_i(z)$  are  $|z| > 3$  and  $|z| < 3$ . In both cases, the region of convergence overlaps with  $|z| > 2$ .

$$\text{If } |z| > 3 \rightarrow h_i[n] = \frac{7}{9}\delta[n] - \frac{2}{3}\delta[n-1] + \frac{2}{9}(3)^n u[n]$$

$$\text{if } |z| < 3 \rightarrow h_i[n] = \frac{7}{9}\delta[n] - \frac{2}{3}\delta[n-1] - \frac{2}{9}(3)^n u[-n-1]$$

long division:

1) Start med at arrangere efter lavest eksponent.

Taller / numerator:  $1 - 3z^{-1} + 2z^{-2} \rightarrow 2z^{-2} - 3z^{-1} + 1$

nevner / denominator:  $1 - 3z^{-1} \rightarrow -3z^{-1} + 1$

2) opstil på long division

$$-3z^{-1} + 1 \overline{) 2z^{-2} - 3z^{-1} + 1}$$

3) Hvor mange gange går

$-3z^{-1}$  op i  $2z^{-2}$ ? resultatet

af dette, skrives her

$$-\frac{2}{3}z^{-1} \overline{) 2z^{-2} - 3z^{-1} + 1}$$

4) Nu ganges  $-\frac{2}{3}z^{-1}$  med

$-3z^{-1} + 1$ , og resultatet

skrives under "Dividend"

$$\begin{array}{r} -\frac{2}{3}z^{-1} \\ -3z^{-1} + 1 \overline{) 2z^{-2} - 3z^{-1} + 1} \\ \hline 2z^{-2} - \frac{2}{3}z^{-1} \end{array}$$

5) Nu trækkes  $(2z^{-2} - 3z^{-1} + 1) - (2z^{-2} - \frac{2}{3}z^{-1})$

Resultatet skrives under dem.

$$\begin{array}{r} -\frac{2}{3}z^{-1} \\ -3z^{-1} + 1 \overline{) 2z^{-2} - 3z^{-1} + 1} \\ \hline 2z^{-2} - \frac{2}{3}z^{-1} \\ \hline -\frac{7}{3}z^{-1} + 1 \end{array}$$

6) Hvor mange gange går

$-3z^{-1} + 1$  op i  $-\frac{7}{3}z^{-1} + 1$

Resultatet skrives oppe i toppen

$$\begin{array}{r} -\frac{7}{3}z^{-1} + \frac{7}{9} \\ -3z^{-1} + 1 \overline{) 2z^{-2} - 3z^{-1} + 1} \\ \hline 2z^{-2} - \frac{7}{3}z^{-1} \\ \hline -\frac{7}{3}z^{-1} + 1 \end{array}$$

7)  $\frac{7}{9}$  ganges med  $-3z^{-1} + 1$

Resultatet skrives her

$$\begin{array}{r} -\frac{7}{3}z^{-1} + \frac{7}{9} \\ -3z^{-1} + 1 \overline{) 2z^{-2} - 3z^{-1} + 1} \\ \hline 2z^{-2} - \frac{7}{3}z^{-1} \\ \hline -\frac{7}{3}z^{-1} + 1 \\ -\frac{7}{3}z^{-1} + \frac{7}{9} \end{array}$$

8) Sidst men ikke mindst

$$\left(-\frac{7}{3}z^{-1} + 1\right) - \left(-\frac{7}{3}z^{-1} + \frac{7}{9}\right)$$

Resultatet skrives under

$$\begin{array}{r} -\frac{7}{3}z^{-1} + \frac{7}{9} \\ -3z^{-1} + 1 \overline{) 2z^{-2} - 3z^{-1} + 1} \\ \hline 2z^{-2} - \frac{7}{3}z^{-1} \\ \hline -\frac{7}{3}z^{-1} + 1 \\ -\frac{7}{3}z^{-1} + \frac{7}{9} \\ \hline \frac{2}{9} \end{array}$$

Dette er den nye taller